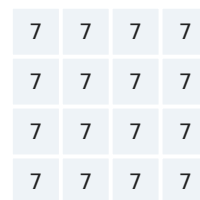
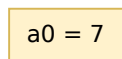


One instruction per row. Colour tracks where data goes (numbers where the value matters).
Flipping CSR 0x7c0 rotates each spatial op 90° so a pixel is free to move anywhere on the grid.
(vadd / vmv / vmerge are element-wise, so H = V.)

Copy one scalar into every lane.

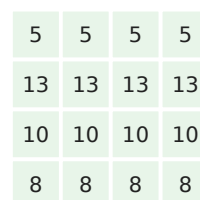
Horizontal mode Vertical mode
(0x7c0=1)



= same ($H = V$)

every lane = 7

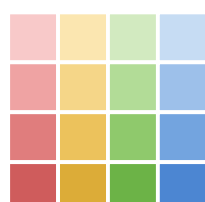
Element-wise add.
Position never
moves.



= same ($H = V$)

 $A + B$

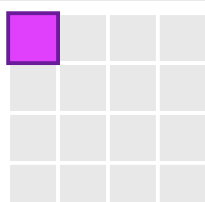
Shift by one. Mode picks the axis.



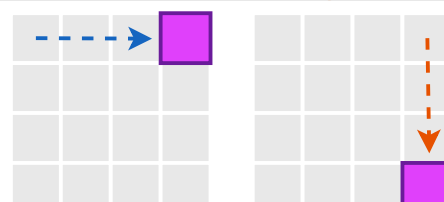
shift RIGHT (within row)

shift DOWN (between rows)

Read any lane $\rightarrow$ any lane. Pixels are not local!



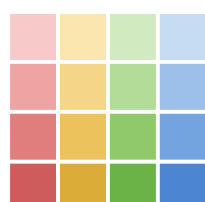
pixel at top-left



H: top-left $\rightarrow$ top-right

V: top-right $\rightarrow$ bottom-right

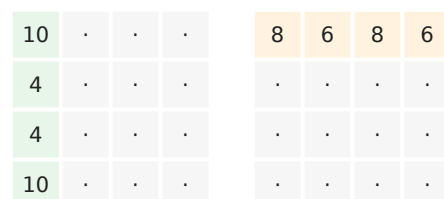
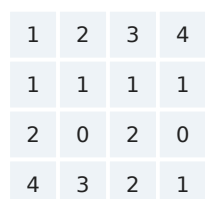
Vertical load/store
transposes the
image.



H: copy through

V: transpose (rows $\leftrightarrow$ cols)

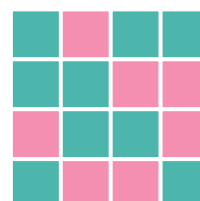
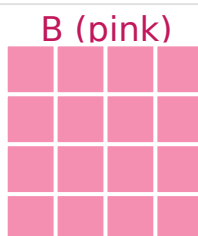
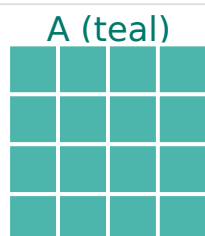
add up. Mode picks the axis the sum lands on.



H: row-sums in column 0

V: col-sums in row 0

v0
Predicate per lane.
Pick A where
mask=1, else B.



*mask gates the
SAME lanes in H
& V; only the
data orientation
under it changes*

A where mask=1, else B